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Microaerated UASB reactor treating textile wastewater: The core microbiome and removal of azo dye Direct Black 22

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HIGHLIGHTS

- Microaeration in UASB reactor promoted complete removal of azo dye *Direct Black* 22.
- Similar COD and sulfate removal for anaerobic and microaerated UASB reactors.
- Effluent toxicity was removed after microaeration.
- Putative aromatic amines degraders microbes developed in the microaerobic zone.

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ABSTRACT

Sequential anaerobic and aerobic processes have been recommended to treat textile wastewater reliably. In this work, the focus was on finding an energetically more competitive system to remove tetra-azo dye Direct Black 22 (DB22). We operated two upflow anaerobic sludge blanket (UASB) reactors (R1 and R2) in three phases (PI, PII, and PIII). R1 was operated as a conventional UASB, while R2 was microaerated in the upper part (0.18 ± 0.05 mg O_2. L^{-1}), aiming to remove DB22 simultaneously with the aromatic amine byproducts. PI consisted of feeding reactors with synthetic textile wastewater (STW), PII had higher salinity in the STW, and PIII was the same as PII, plus sulfate. The results showed that color and COD removal efficiencies were similar for both reactors (67–72% for R1 and 59–78% for R2) without a substantial influence of oxygen in R2. However, microaeration played a crucial role in R2 by removing the anaerobically formed aromatic amines; during PIII, the effluent was 16 times less toxic than that of R1. The microbial community that developed in the sludge bed of both reactors was quite similar, with the core microbiome represented by *Trichococcus, Syntrophus and Methanosaeta* genera. The increase in salinity in PII and PIII promoted a shift in the microbial community, excluding salty-sensitive genera from the core microbiome. The putative genera *Brevundimonas* and *Ornatilinea* were associated to aromatic amine microaerobic removal. Therefore, there is a potential application of a compact microaerated anaerobic system for textile wastewater treatment.

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1. Introduction

Textile and apparel industries are a significant sector of the national economy in low-middle income countries (GNI < $12,055 – The World Bank, 2017). The faster this industrial sector grows, the faster the environmental issues spread due to the resulting pollution of its activities. Global clothing production has been developed mainly in Bangladesh, India, Korea, Cambodia, Brazil and Pakistan (Tewari, 2005), and China, which alone is responsible for 32% of the textile fabrics and production of ready-made garments in the world (AEPC, 2018). Unskilled workers and poor infrastructure are still common features of the textile industries in most of those countries. Additionally, the lack of regulation associated with the use of a huge variety of dyes and chemicals contribute to increased pollution, making it difficult to standardize treatment processes for textile wastewater.

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Synthetic dyes are industrially preferred over natural dyes due to their high fixation properties. The dyes mostly used in industries, such as textile, cosmetics, and paper printing, are synthetic azo dyes corresponding approximately to 70% of the total dyes used by these industries (Stick and Mocerino, 2009). During the dyebath in textile industries, fixation to fabrics is usually incomplete, resulting in wastewater very rich in dyes. An advantageous eco-friendly and low-cost treatment alternative is the use of two-step biological processes. The first step consists of an anaerobic phase in which the azo bonds are cleaved by azoreductase enzymes, resulting in colorless wastewater, although full of toxic byproducts (aromatic amines) (Field et al., 1995; Weisburger, 2002). A further aerobic step is required to achieve the bioconversion or complete removal of the aromatic amines (Van Der Zee and Villaverde, 2005; Bonakdarpour et al., 2011; Khan et al., 2014; Sandhya et al., 2005).

Sequencing batch reactors (SBR) and biofilm electrode reactors (BER), operating under microaerobic or anaerobic-aerobic conditions, have efficiently been used for textile wastewater treatment at the bench scale, removing from 70% to 90% of COD and color, respectively (Cao et al., 2018; Çinar et al., 2008; Srinivasan and Sadasivam, 2018). However, although promising reactor setups, they have not been sufficiently tested on a real scale. Additionally, those reactor configurations demand skilled operators due to the high level of automation applied and frequent maintenance requirements, thus hampering their application for many small and moderately sized industries. In contrast, the UASB reactor type is well known for its easy handling and low operational costs (Lettinga and Pol, 1991); comparatively, it also has high efficiency for azo dye decolorization (Patel, 2015). UASB has also shown good efficiency in real textile wastewater treatment in a pilot-scale reactor (Amaral et al., 2014). However, the effluent still demands an aerobic post-treatment to guarantee aromatic amine removal and, consequently, remove wastewater toxicity.

Therefore, in this work, a novel approach system to treat textile wastewater was evaluated using microaeration in a conventional UASB. The aim was to create two different redox zones: an anaerobic zone deeper in the sludge bed and a microaerobic zone from the middle to the upper part of the reactor. The complete degradation of azo dyes in a single treatment unit was the main target of the approach. We also investigated the effects of microaeration, salinity, and sulfate on the structure of the microbial community. The performance of the reactor was based on the chemical oxygen demand (COD) and color removal efficiency, as well as the resulting effluent toxicity.

2. Experimental procedures

We operated two identical UASB acrylic reactors (R1 and R2 – Fig. 1) with a 6 L working volume (12 cm diameter and 60 cm height) under mesophilic temperature (25.0 ± 1.4 °C). Sampling ports were placed at 4 (S4), 14 (S14), and 24 cm (S24) from the bottom of the reactors. R1 was a traditional UASB reactor and operated as a control. R2 was partially microaerated by an air diffuser placed 25 cm from the bottom and supplied by an aquarium pump (Boyu®, U-2800, China). The dissolved oxygen (DO) was kept at 0.18 ± 0.05 mg O_2 L^{-1} at sampling port S24.

We fed the reactors with synthetic textile wastewater (STW), which was refrigerated (7 ± 1 °C) and pumped (model 323, Watson Marlow®, United Kingdom) to the reactors, passing through a water bath (40 ± 5 °C) (NI1225, Nova Instruments®, Brazil). The mesophilic reactor temperature range was maintained using a heating jacket (thermal blanket, Etaflon®, Brazil). Both reactors were inoculated with anaerobic sludge (2.62 g STV L^{-1} in the mixed liquor) from a pilot-scale UASB reactor that was previously used to treat real textile wastewater (Amaral et al., 2014).

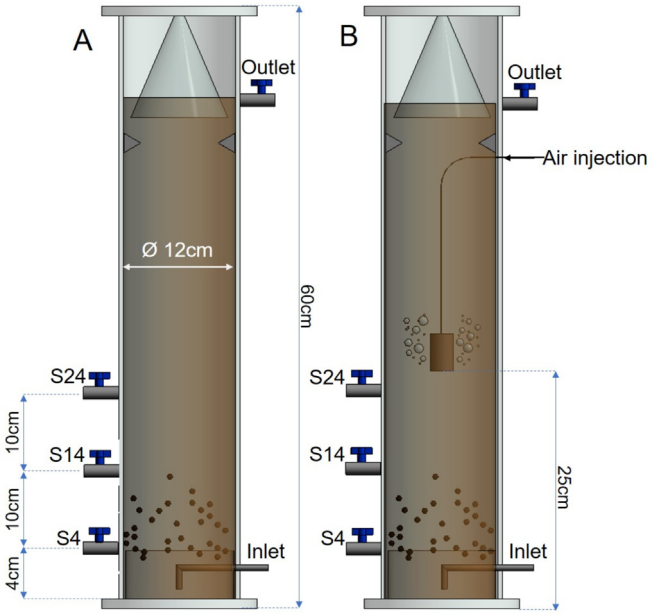


Fig. 1. Schematic illustration of the two acrylic reactors: (A) conventional UASB - R1 and (B) microaerated UASB - R2.

We conducted the experiment for over 290 days divided into three operational phases (P). At phase PI (150 days the STW contained natural salinity of 1.9 g NaCl L^{-1}, while at phase PII (100 days) sodium chloride was added to STW to reach salinity of 2.9 g NaCl L^{-1}, and at PIII (40 days) salinity was maintained, as in PII, plus extra sulfate of 250 mg SO_4^{2-} L^{-1}.

2.1. Synthetic textile wastewater (STW)

The STW was based on a real textile wastewater composition, as described by Amorim et al. (2013). It was prepared using 0.06 mM of the tetra-azo dye Direct Black 22 (DB22, $C_{44}H_{32}N_{13}Na_3O_{11}S_3$; C. I. 35435; CAS 6473-13-8; molecular weight of 1083.97 g mol^{-1}); starch (1200 mg O_2 L^{-1} as COD); and macro and micronutrient solutions (Florencio et al., 1993). The dye was prehydrolyzed before use by adjusting the pH to 11.0 ± 0.05 using NaOH (20% w:w), followed by 1 h heating at 80 °C and a final pH readjustment to 7.0 ± 0.05 at room temperature using HCl (37%). We prepared the starch solution using commercial corn starch (Maizena®, Brazil) and tap water by stirring at 90 °C for 30 min.

2.2. Reactor monitoring

The reactors were monitored based on the following parameters: flow rate, pH, temperature, and DO (daily); color, chemical oxygen demand (COD), oxidation-reduction potential (ORP), alkalinity, and sulfate (3 times a week); and sulfide (2 times a week). The analytical methods followed the *Standard Methods for the Examination of Water and Wastewater* (APHA, 1998). The aromatic amine formation was qualitatively evaluated through spectrophotometric light scanning at the range of 200–350 nm wavelength, as recommended by Pinheiro et al. (2004), using filtered samples (fiberglass membrane 0.45 μm, Sartorius®, Germany).

2.3. Toxicity analysis

The assays were performed using *Vibrio fischeri* as a toxicity indicator microorganism following the protocols of Microtox®

(Microtox Model 500 analyzer, Strategic Diagnostics, Inc. SDIX, Newark-DE, USA). We monitored cell luminescence inhibition during 30 min. The results were expressed as the dilution factor (DF) required for the effluent not to cause toxic effects on *Vibrio fischeri*.

2.4. Identification of the microbial community

We collected eight biomass samples for DNA extraction: inoculum (sample 1); sludge bed of R1 at the end of each operational phase (sample 2 – PI; sample 3 – PII; and sample 4 – PIII); sludge bed of R2 at the end of each operational phase (sample 5 – PI; sample 6 – PII; and sample 7 – PIII); and sludge attached to the air diffuser in R2 at the end of PIII (sample 8). We used a Power Kit Soil® (MO Bio laboratories, Carlsbad-CA, USA) for DNA extraction according to the manufacturer's protocols. DNA quantification was carried out using a NanoDrop2000 spectrophotometer (Thermo Scientific, USA). The DNA was stored at −20 °C and was subjected to amplicon sequencing (bTEFAP®, USA) using Illumina MiSeq. Sequencing was performed by Mr. DNA Molecular Research LP (Shallowater-TX, USA) using the universal primer 515 F paired with 907R for both domains, *Achaea* and *Bacteria*, with 20,000 reads and 2 × 300 bp. The pipeline used followed the MiSeq manufacturer's guidelines. Sequences were aligned and depleted of barcodes. Then, sequences with less than 150 bp were discarded, as well as sequences with ambiguous base calls. We clustered the sequences and removed the chimeras. Operational taxonomic units (OTUs) were defined by clustering at 3% divergence (97% similarity). Final OTUs were taxonomically classified using BLASTn against a curated database derived from RDPII and NCBI platforms (www.ncbi.nlm.nih.gov, http://rdp.cme.msu.edu). We deposited the sequencing data yielded in this study in the European Nucleotide Archive under the accession number PRJEB33520.

2.5. Statistical analysis

Statistical Analysis of Metagenomic Profiles (STAMP) (Parks et al., 2014) was the software used to identify the significant changes in community structure that resulted from salinity and sulfate concentration additions over the operational phases. When the p-value was lower than 0.05, the differences were considered statistically significant.

3. Results and discussion

3.1. COD and color removal

The organic loading rate (OLR) ranged from 1.27 to 1.50 kg COD m^{-3} d^{-1} during the entire operational period. The performance of both reactors concerning the COD removal was similar for each operational phase (Table 1). In PI, the removal efficiencies for R1 and R2 were 72 ± 17% and 78 ± 18%, respectively; in PII, 67 ± 10% and 59 ± 10%; and in PIII, 69 ± 6% and 70 ± 8%. These results show first that the oxygen supplied to R2 did not enhance the removal efficiency ($p > 0.05$). Second, by increasing the salinity level from 1.9 (PI) to 2.9 g of NaCl L^{-1} (PII), we observed a decrease in the removal efficiency for both reactors. However, in the case of R1, the difference was not statistically significant (p 0.6), while for R2, the removal efficiency dropped from 78% (PI) to 59% (PII) ($p = 0.001$). Third, recovery was observed from PII to PIII in both reactors, but especially for R2 (from 59% to 70%); this can be an indication of microbial adaptation to the higher salinity level associated with the sulfate added, applied in PIII. In PII, the drop in COD removal efficiencies observed for both reactors can be attributed to the changes in the microbial community (data shown on item 3.4), which was affected by the higher salinity compared to that of PI.

Table 1
COD concentrations of influent and effluent of the UASB conventional reactor R1 and the microaerated reactor R2 during the three operational phases.

Operational phase	COD (mg O_2 L^{-1})		
	Influent	R1 effluent	R2 effluent
PI	1,334 ± 340 mg L^{-1}	387 ± 259 mg L^{-1}	306 ± 282 mg L^{-1}
PII	1,530 ± 264 mg L^{-1}	495 ± 172 mg L^{-1}	603 ± 205 mg L^{-1}
PIII	1,534 ± 284 mg L^{-1}	469 ± 64 mg L^{-1}	442 ± 74 mg L^{-1}

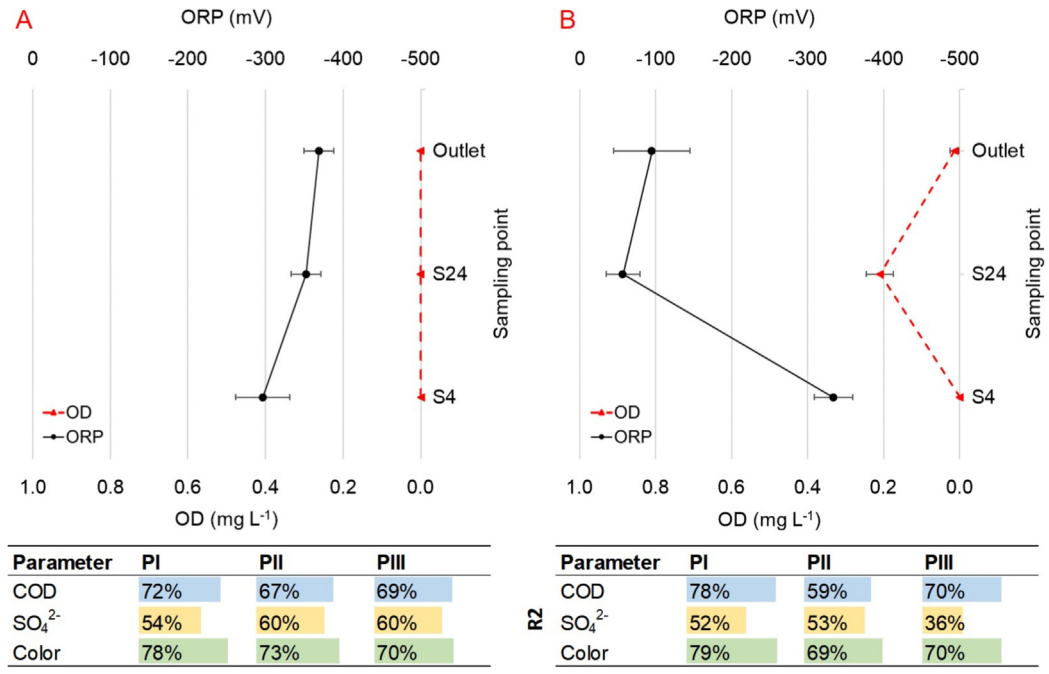
The recovered COD removal efficiency in PIII might be related to sulfate addition because sulfate-reducing bacteria (SRB) use electrons from the organic matter more efficiently than methanogens due to the thermodynamic advantage (Muyzer and Stams, 2008), competing mainly by propionate and butyrate.

Concerning the color removal, the performance was also similar for both reactors in the first two phases. In PI, the removal efficiency was 78 ± 10% for R1 and 80 ± 12% for R2. In PII, the values were slightly lower, 73 ± 17% for R1 and 71 ± 11% for R2. Comparing these results with those of COD removal, there is a noticeable difference in the salinity influence on the microorganisms. The higher salinity in PII compared to that in PI resulted in a decrease of color and COD removal efficiencies, especially for R2. However, in the case of color removal, the R2 efficiency decreased by only 9%, while that of COD removal was 19%. One explanation is that dye-degrading microorganisms are less sensitive than those of COD.

The influence of the added sulfate on PIII color removal efficiency (69% ± 9% for R1 and 69 ± 11% for R2) was not significant when comparing the results of both reactors ($p = 0.97$). Color removal may either be stimulated or suppressed by sulfate because it competes with azo dyes for the electrons available, and sulfide also stimulates dye chemical removal (Prato-Garcia et al., 2013). Simultaneous sulfate and azo dye reduction are also supported by the ORP, which was − 300 ± 86 mV in the sludge bed of both reactors during PIII (S4, Fig. 2). Lourenço et al. (2001) reported that color removal efficiency increased with lower ORP values. Sulfate reduction occurs when ORP is approximately −220 mV, considering SO_4^{2-} degradation over H_2S formation (Yoo et al., 2001), whereas azo dye reduction is reported to occur in the range from −50 mV to −600 mV depending on the type of azo dye (Dos Santos et al., 2004; Rau and Stolz, 2003).

3.2. Sulfate removal

In PIII, the sulfate removal efficiency was 60 ± 10% and 36 ± 17% for R1 and R2, respectively (Fig. 2). The low R2 efficiency can be attributed to the microaeration supplying the futile sulfur cycle. In the bottom part of R2, the influent sulfate was anaerobically converted to sulfide, which was partially reconverted to sulfate under oxygen exposure in the upper part. The ORP profile indicated that anaerobic conditions were established across the whole R1 height (ORP from −296 to −368 mV - Fig. 2A), while typical microaerobic ORP was detected in R2 at the height S24 (−56.7 ± 22 mV) and at the outlet (−95.1 ± 50 mV). Thus, favorable conditions were confirmed in R2 for the sulfur futile cycle establishment. From a microbiological perspective, the sulfate removal efficiency decrease in R2 can also be attributed to the high sensitivity of SRB microbes to molecular oxygen as they are strictly anaerobes. Oxygen inactivates various enzymes of SRB when exposed under long periods (Fukui, 2002; Krekeler et al., 1998).



R1	Parameter	PI	PII	PIII
	COD	72%	67%	69%
	SO_4^{2-}	54%	60%	60%
	Color	78%	73%	70%

R2	Parameter	PI	PII	PIII
	COD	78%	59%	70%
	SO_4^{2-}	52%	53%	36%
	Color	79%	69%	70%

Fig. 2. Height ORP profiles and removal efficiencies of COD, sulfate and color during the three operational phases for R1 (A) and R2 (B).

3.3. Aromatic amines

We detected aromatic amines in the effluent of both reactors only after the first 30 days of operation (data not shown). Their removal was observed only in the microaerated R2 from day 148 onwards. In R1 during the entire experiment, the formed aromatic amines were not removed, and they were always detected in all three sampling points and the reactor effluent (Fig. 3). In R2, the aromatic amines that formed in the anaerobic zone, detected at sampling points S4 and S14, were partially removed from R2 effluent (Fig. 3).

Concerning the toxicity on *Vibrio fischeri*, two factors were considered in the STW, the azo dye itself and the salinity. The acute toxicity endpoint is the effective concentration of a chemical that causes a 50% reduction in the bioluminescence of the bacteria. We did not identify significant acute toxicity in the azo dye DB22-containing STW (Table 2), as also reported by Menezes et al.

Table 2
Toxicity level of *Vibrio fischeri* measured as a dilution factor of the STW and effluent of reactors R1 and R2 at the end of each operational phase.

Operational phase	Dilution factor		
	STW	R1 effluent	R2 effluent
PI	1	2	1
PII	2	14	1
PIII	2	16	1

DF > 1 indicates the acute toxicity level for *Vibrio fischeri*.
DF means how many dilutions are needed to dilute the influent/effluent to prevent acute toxicity.

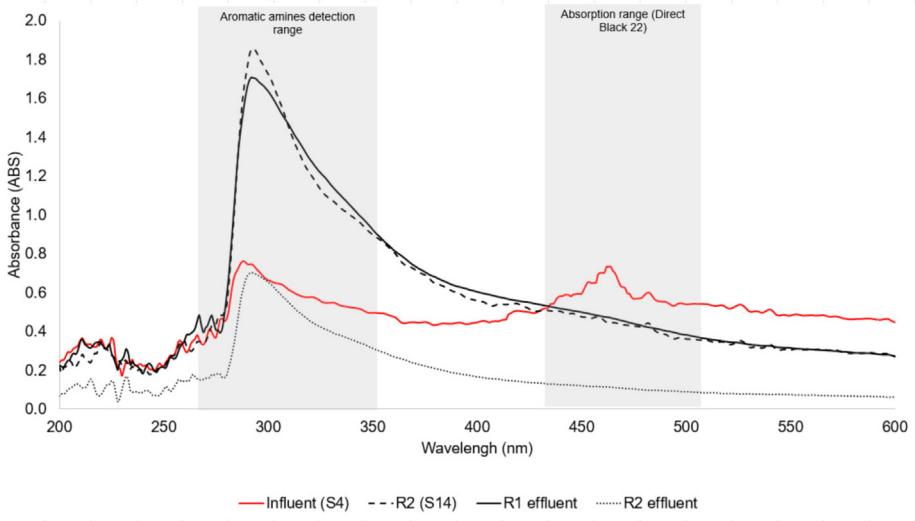


Fig. 3. UV/VIS scan performed on day 198 (during PII) showing absorbance peaks of aromatic amine formation and removal.

(2019). Low permeability of azo dye for cells due to the presence of sulfonated groups in the DB22 molecular structure may be responsible for the absence of acute toxicity. In the case of the salinity addition effect, the STW dilution factor increased from 1 (PI) to 2 (PII and PIII), indicating salt toxicity against *Vibrio fischeri*, which is known to be affected by salinity (Cook et al., 2000). In the R1 effluent, aromatic amines were associated with an increased and strong toxicity level, as shown by a dilution factor of 14 and 16, measured at the end of PII and PIII, respectively. On the other hand, in the R2 effluent, the aromatic amines were removed or converted into nontoxic compounds, resulting in the absence of acute toxicity. Therefore, these results provide evidence of the microaeration role on aromatic amines removal, complementing the anaerobic process for azo dyes removal.

3.4. Microbial community

The most abundant phyla in the sludge bed samples from both reactors and their respective relative abundance (RA) range during the experiment were *Proteobacteria* (13.17–44.21%), *Firmicutes* (7.12–41.94%), *Bacteroidetes* (13.05–26.17%), and *Chloroflexi* (2.51–4.77%). These phyla are commonly found in the microbial community of anaerobic systems, usually with RA above 1% (Köchling et al., 2017; Okada et al., 2013).

Thirty-six OTUs were identified in R1 sludge bed samples at the genus level (with RA> 1% at least in one of the phases), of which 17 were prevalent with RA higher than 0.1% during the entire experiment (Fig. 4). In the case of R2 samples, we identified a total of 31 OTUs, of which 15 were also considered to be of high prevalence (RA>1%).

In this work, the COD removal in both reactors (Table 1) was supported by the core microbiome related to carbon cycling because it was composed of the fermentative bacterial genera of *Trichococcus*, *Bacteroides*, *Clostridium,* and *Fusobacterium*, methanogenic archaea genera of *Methanosaeta* and *Methanobacterium,* and an important syntrophic bacterial genus, i.e., *Syntrophus*. The latter is responsible for metabolic linking of fermentative and methanogenic microbes. Syntrophic interactions can support anaerobic digestion from organic compounds to methane (Jackson et al., 1999). The syntrophic chain was well established over the operational phases in R1 and R2 with increasing RA of the fermentative *Trichococcus*, from less than 1% and 2.45% in PI to 7.51% and 16% in PII and to 11.58% and 34.58% in PIII, respectively. This genus might have used starch as a substrate to produce acetate, formate, and ethanol. Following the chain, the strict anaerobe *Syntrophus* also exhibited increasing RA from PI (<1% in R1 and R2) to PIII (4.27% in R1 and 3.29% in R2). These species are known as oxidizers of volatile fatty acids, benzoate, and crotonate to acetate in the presence of H_2/formate-utilizing methanogens (Kuever, 2014; Jackson et al., 1999). The occurrence of the *Methanobacterium* genus, a H_2/formate-utilizing archaea, supported *Syntrophus* activity. Finally, acetate was converted to methane by *Methanosaeta*, where RA reached 8.50% and 2.61% in PIII, for R1 and R2, respectively. The lower RA found in R2 compared to R1 can be explained by the possible oxygen sensitivity of some methanogens (Kiener and Leisinger, 1983; Patell and Sprott, 1990). Menezes et al. (2019) detected a 50% inhibition of the methane production rate when operating a batch reactor with intermittent microaeration and similar wastewater composition, except for sulfate and salinity.

The coexistence of many other metabolisms in the core microbiome was due to the wastewater composition because aromatic (*Pseudomonas, Flavobacterium, and Sphingobacterium*) and sulfate-reducing microbes (*Desulfobulbus, Desulfovibrio*), for instance, were also identified. All these genera play a key role during the treatment of textile wastewater.

The main sulfate-reducing bacterial (SRB) genera identified in the microbial community from both reactors were *Desulfobulbus, Desulfomicrobium, Desulfotomaculum,* and *Desulfovibrio.* In R1, the RA of all SRB species increased over the operational phases, corresponding to 4.07% (PI), 6.33% (PII) and 8.91% (PIII). In R2 however, the RA of all SRB species slightly decreased (5.64% in PI, 3.46% in PII, and 3.78% in PIII), which supports the hypothesis that microaeration affected SRB development.

The beta diversity of the OTUs was analyzed using the Bray-Curtis index (Fig. 5). We observed the first shift in the microbial community by comparing the microbiome identified in the

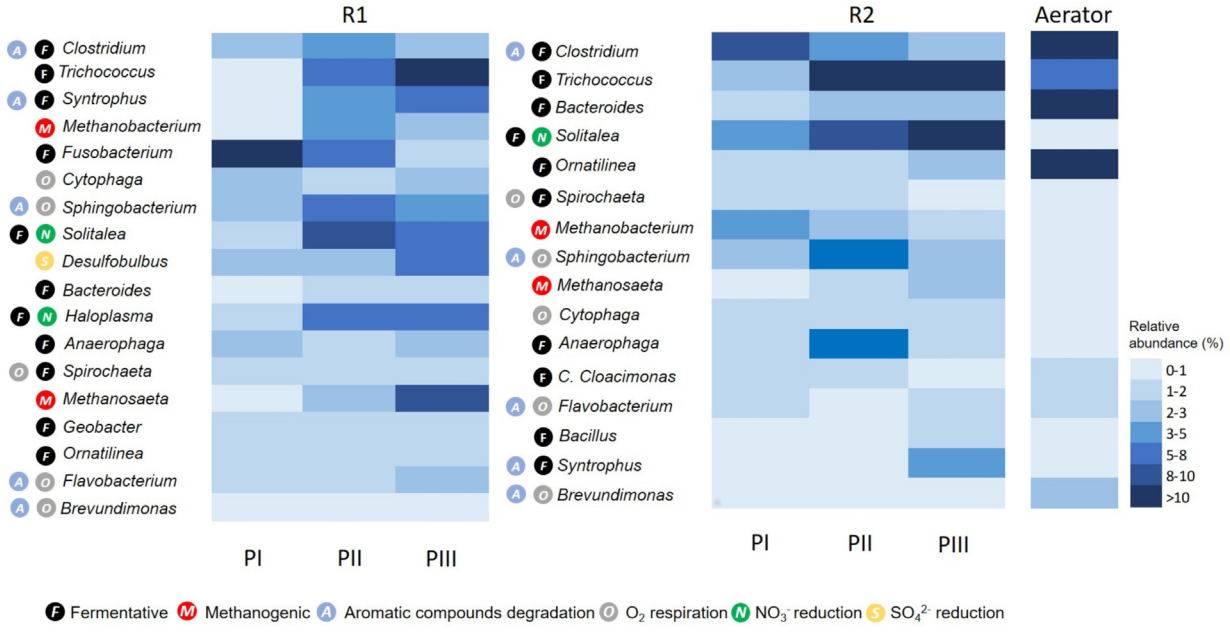


Fig. 4. Heatmap representing the core microbiome (RA of 0.5% or higher) and its known metabolic capabilities at the genus level.

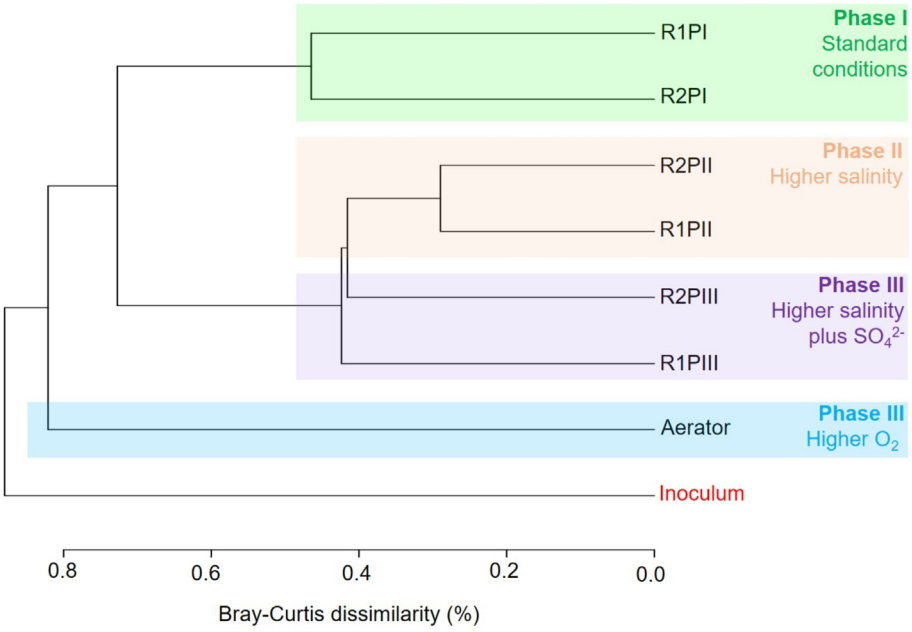


Fig. 5. Dendrogram showing the taxonomic divergence at species level, highlighting the differences in STW for PI, PII and PIII.

inoculum (day 1 of operation) and the microbiome identified at the end of PI (day 150 of operation). In that case, R1 and R2 were 87% phylogenetically divergent from the inoculum (Fig. 5). However, the diversity among R1 and R2 in PI was quite similar (48% similarity, Fig. 5). The microaeration in R2 stimulated the growth of facultative anaerobic genera such as *Pseudomonas* (RA 10.83%) and *Clostridium* (RA 8.79%), while in R1, the presence of obligate anaerobic genera was widely observed, with *Fusobacterium* being the most abundant genus identified in PI (RA 18.62%). Despite the difference in taxonomic structure, a similar metabolic behavior is inferred for the microbial communities of both reactors.

Another important factor is that the first shift in PI was driven by the difference in the composition of the wastewater that was feeding reactors and sludges. The STW used to feed the reactors contained starch and only one type of azo dye, while the real textile wastewater to which the inoculum was previously exposed contained a mixture of dyes, high sulfate concentration, and many chemical compounds as described by Amaral et al. (2017). The genera *Kosmotoga*, *Synergistes, Actinomyces* and *Caldisericum* related to the reduction of sulfur-complexed metals and thiosulfate were identified with RA>1% in the inoculum sample. However, at the end of PI, these genera were not significant in the core microbiome, presenting RA close to 0% during the rest of the experiment.

The salt increase in PII caused the second shift, clustering R1 and R2 with an even closer index of diversity (72% similarity), highlighting the strong influence of salinity on the microbes' selection (higher salinity, Fig. 5). The species *Sulfuricurvum kujiense* was washed out from both reactors due to the salinity increase, even with the availability of its preferable substrates, with RA dropping from 7.49 (PI) to 0.18% (PII) in R1 and from 4.76 (PI) to 0.09% (PII) in R2. This sulfur-oxidizing bacterium utilizes sulfide, elemental sulfur, thiosulfate and hydrogen as electron donors under microaerobic or anaerobic conditions, but it is very sensitive to salinity (Han et al., 2012; Kodama and Watanabe, 2004). On the other hand, some genera endowed with high salinity resistance took the opportunity to increase their dominance in the microbial community. For example, in R2, *Trichococcus pasteurii,* a microbial species associated with carbon removal, exhibited an RA 6.5 times higher in PII (15.86%) and 14 times in PIII (34.58%) than in PI (2.43%). Similar behavior was found for this species in R1, but with lower RA. Sulfate addition in PIII did not cause any significant shift in the microbial beta diversity (Fig. 5).

An important aspect concerning the beta diversity index analysis is related to the microbial community that developed attached to the aerator, which presented only 17% similarity compared with the sludge bed samples. The microorganisms in the reactor zone near the air diffuser, located 25 cm from the reactor bottom, were exposed to oxygen and dye degradation byproducts (i.e. aromatic amines). Therefore, microaerating the intermediate zone of the UASB reactor promoted aromatic amine degradation during PII and PIII. Concerning the developed microbial community, species of the genus *Brevundimonas* were found in the aerator biomass in PIII at an RA of 2.65%. An important highlight related to the *Brevundimonas* genus is that the highest RA in the sludge bed samples of R2, during PI, was 0.26%, while that in R1 was only approximately 0.04%. This genus was studied by Konopka et al. (1989), who identified the ability of some species to mineralize aniline via *meta*-cleavage of the aromatic ring induced by the catechol 2,3 oxygenase enzyme. We identified some other genera related to the degradation of aromatic compounds in the biomass collected in the microaerated zone. The genus *Ornatilinea* was identified with RA of 13.47% in the aerator sample, higher than the RA identified in the sludge bed samples (1.14–2.61%). The fermentative genus *Ornatilinea* is classified within the family *Anaerolineaceae,* which has been commonly identified in mixed cultures of anaerobic reactors, as putative aromatic degrader microorganisms (Kümmel et al., 2015; Zhang et al., 2017; Rosenkranz et al., 2013). Sun et al. (2015), identified a bacterial phylotype within the family *Anaerolineaceae*, demonstrating the capability to assimilate aniline, an aromatic amine potentially formed after azo dye breakdown. The literature has shown important findings about the putative genera *Brevundimonas* and members from the family *Anaerolineaceae*. Nevertheless, further investigation of their enzymatic capabilities is required to expand the current knowledge about these genera in aromatic amines degradation.

4. Conclusions

Microaerating the upper part of a UASB reactor favored the degradation of the tetra-azo dye Direct Black 22. Aromatic amines formed in the lower anaerobic zone were partially removed in the upper microaerobic zone. A large variety of genera able to perform the complete anaerobic digestion of organic matter, while simultaneously removing dye and reducing sulfate were present in the microbial community. The sludge bed core microbiome of both reactors was quite similar, represented by *Trichococcus*, *Syntrophus*, and *Methanosaeta*. Changes in the salinity level and sulfate concentration in the synthetic textile wastewater modified the microbial community structure. However, after adaptation, COD and color removal efficiencies remained high, within the range of 59–78% and 69–79%, respectively, during the operational period. The sulfate removal efficiency in the microaerated UASB reactor diminished during the last operational phase (PIII), dropping from 53% to 36%, reflecting oxygen influence on SRB development. The aromatic amines removal through microaeration produced an effluent free of toxicity, thus confirming its great potential application.

Declaration of competing interest

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